

CH 3

Q.1.What is learning from examples in machine learning?

In machine learning, "learning from examples" refers to the process where a computer model learns patterns or knowledge by analyzing data that has already been labeled or categorized. This is similar to how humans learn by seeing examples or experiences.

Here's how it works in simple terms:

1. **Training Data:** You provide the machine with a set of examples, which is called the *training data*. These examples have both the input (features) and the correct output (label or result).
2. **Learning:** The machine uses this training data to find patterns or relationships between the input (e.g., features like height, weight, age) and the output (e.g., the label like "sick" or "healthy").
3. **Making Predictions:** Once the machine has learned from these examples, it can make predictions or decisions on new, unseen data by recognizing similar patterns.

For example, if you are teaching the machine to recognize whether an email is "spam" or "not spam," you give it examples of emails labeled as spam or not spam. The machine then learns the differences (like certain words or email structures) and uses this knowledge to classify new emails.

In short, "learning from examples" is how machine learning algorithms learn by studying labeled data to make predictions on new data.

Q.2.What is supervised learning? Provide an example.

Supervised Learning is a type of machine learning where the model is trained using labeled data. In supervised learning, the algorithm learns from the input-output pairs (examples) you provide. The goal is for the model to learn the relationship between the input data and the correct output so that it can predict the output for new, unseen data.

Key Points:

- **Labeled Data:** You give the model a dataset where each input (feature) is paired with the correct output (label).
- **Training:** The algorithm learns by comparing its predictions to the correct output and adjusting to make better predictions.
- **Prediction:** After learning, the model can make predictions on new data

based on the patterns it has learned.

Example:

Imagine you want to teach a machine to predict whether a student will pass or fail an exam based on how many hours they studied.

- **Input (Features):** Hours studied by the student.
- **Output (Label):** Whether the student passed or failed.

Training:

You provide the machine with examples of students who studied for different hours and their corresponding results (pass/fail). For example:

Hours Studied	Exam Result
1	Fail
3	Fail
5	Pass
7	Pass

Learning:

The machine learns the relationship between hours studied and exam result. It notices that the more hours a student studies, the higher the chance of passing.

Prediction:

Now, if you give the machine a new student who studied for 4 hours, it might predict that the student will fail based on its learned pattern (since 4 hours is closer to the "fail" examples in the training data).

In Summary:

Supervised learning is like teaching a model with examples, and it helps the model predict outcomes for new data based on what it learned from previous labeled examples.

Q.3. Define unsupervised learning. How does it differ from supervised learning?

Unsupervised Learning is a type of machine learning where the model is trained using data that does **not** have labels or specific output values. In unsupervised learning, the goal is to find hidden patterns or structures in the data without being told what the correct answers are.

Key Points:

- **No Labels:** In unsupervised learning, you only provide the input data, and there is **no predefined output** or label for the data.
- **Learning:** The model tries to discover patterns, groupings, or relationships within the data on its own.
- **Clustering and Association:** The model might group similar data

together (clustering) or find associations between different data points (association).

Example:

Imagine you have a collection of pictures of animals, but you don't know what animals are in the pictures. Using unsupervised learning, the model might group similar pictures together based on features like size, shape, or color.

For example:

- Group 1: Dogs
- Group 2: Cats
- Group 3: Birds

Even though the model didn't know the names of the animals, it was able to group similar animals together based on patterns it discovered.

Difference Between Supervised and Unsupervised Learning:

Supervised Learning	Unsupervised Learning
Labeled Data: Requires labeled data (input and correct output).	Unlabeled Data: Uses data without labels or predefined outputs.
Goal: The model learns to predict the output for new inputs.	Goal: The model tries to find hidden patterns or structures in the data.
Examples: Classification, regression (e.g., predicting whether an email is spam or not).	Examples: Clustering, association (e.g., grouping customers by their buying behavior).

In Summary:

- **Supervised Learning** is about learning from labeled data to predict outputs.
- **Unsupervised Learning** is about finding hidden patterns or groupings in data without labels.

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Q.4.Explain reinforcement learning and provide a simple example.

Reinforcement Learning (RL) is a type of machine learning where an agent (a model or system) learns by interacting with an environment. The agent takes actions in the environment and receives **rewards** or **penalties** based on those actions. The goal is to maximize the total reward over time by learning which actions lead to the best outcomes.

In simple terms, the agent learns by trial and error, similar to how we learn from experience. The agent tries different actions, sees the results, and adjusts its behavior to get better outcomes.

Key Concepts:

- **Agent:** The system or model that makes decisions.
- **Environment:** The world the agent interacts with.
- **Action:** The decision or move the agent makes in the environment.
- **Reward:** The feedback the agent gets from the environment after taking an action (positive or negative).
- **Policy:** The strategy the agent follows to decide which actions to take based on the state of the environment.
- **Goal:** The agent's objective is to maximize the total reward it receives over time.

Example:

Imagine you are training a dog to fetch a ball. The dog is the agent, and the environment is the park where the dog plays. Here's how reinforcement learning works:

1. **Action:** The dog tries to fetch the ball (action).
2. **Reward:** If the dog successfully retrieves the ball, you give it a treat (positive reward). If the dog does not fetch the ball, you don't give it a treat (negative or no reward).
3. **Learning:** Over time, the dog learns that fetching the ball gives it a treat, so it keeps trying to fetch the ball to get more rewards.

This is similar to how a reinforcement learning agent learns to make decisions in an environment to maximize rewards.

In Summary:

Reinforcement Learning is about learning through rewards and penalties. The agent takes actions, receives feedback from the environment, and learns the best actions to take in order to maximize its rewards.

Q.5.What is linear regression in machine learning?

Linear Regression is a simple machine learning algorithm used for predicting a continuous value based on one or more input features. It tries to find the best-fitting straight line (also called a *regression line*) through a set of data points. The line represents the relationship between the input and the output, helping us predict future values.

Key Points:

- **Input (Features):** These are the independent variables, also called predictors or features. In simple linear regression, there is usually just one feature.
- **Output (Target):** This is the dependent variable, the value that we want

to predict.

- **Equation:** In simple linear regression, the relationship between input and output is represented by the equation:

$$y = mx + b$$

Where:

- y is the predicted output (target),
- x is the input (feature),
- m is the slope of the line (how much y changes for a given change in x),
- b is the y-intercept (where the line crosses the y-axis).
- **Goal:** The goal of linear regression is to find the values of m and b that minimize the difference between the predicted values and the actual data points (this difference is called the *error*).

Example:

Let's say you want to predict the salary of a person based on their years of experience. You have the following data:

Years of Experience	Salary (in \$)
1	30,000
2	35,000
3	40,000
4	45,000

Using linear regression, the algorithm will find the best line that fits this data, and then you can use that line to predict the salary for a person with, say, 5 years of experience.

For example, if the line equation is:

$$y = 5,000x + 25,000$$

Then, for 5 years of experience, the predicted salary would be:

$$y = 5,000(5) + 25,000 = 50,000$$

In Summary:

Linear regression is used to predict a continuous value by fitting a straight line through data points. It works by finding the relationship between input features and the output target, which allows you to make predictions for new, unseen data.

Q.6.What is logistic regression used for in machine learning?

Logistic regression is a machine learning algorithm used for **classification problems**, where the goal is to predict the category or class of an item based on certain input features.

For example, it can be used to predict:

- Whether an email is **spam** or **not spam**.
- Whether a customer will **buy** or **not buy** a product.

In logistic regression:

- The algorithm uses a **mathematical equation** to find a relationship between input features (like age, income, or education level) and the output class (like "yes" or "no").
- The output is a **probability** that a particular input belongs to a certain class. For example, it might say there's an 80% chance that the email is spam.
- The final decision (spam or not spam) is made by setting a threshold. If the probability is above a certain value (like 0.5), it classifies it as "spam," otherwise "not spam."

To summarize in simple terms: **Logistic regression helps to predict categories (like yes/no or 0/1) based on input data, using a mathematical approach to find probabilities.**

Q.7.Explain the different forms of learning in machine learning: supervised, unsupervised, and reinforcement learning.

In machine learning, there are three main types of learning: **Supervised Learning**, **Unsupervised Learning**, and **Reinforcement Learning**. Let me explain each one in simple terms:

1. Supervised Learning:

- **What it is:** In supervised learning, the machine learns from a **labeled dataset**. This means the input data comes with the correct answer (or label).
- **How it works:** The algorithm is trained using both the input data and the correct answers. It tries to learn the relationship between the input and the output so that it can predict the correct label for new, unseen data.
- **Example:**
 - **Email spam filter:** The algorithm is trained using a dataset of emails that are labeled as "spam" or "not spam". It learns the patterns in the emails and can then classify new emails correctly.

- **Image recognition:** The algorithm is trained on images of cats and dogs, each labeled accordingly. Later, it can classify new images as either a cat or a dog.

2. Unsupervised Learning:

- **What it is:** In unsupervised learning, the machine works with data that **does not have labels**. The goal is to find hidden patterns or relationships in the data without knowing the correct answers beforehand.
- **How it works:** The algorithm looks for patterns like grouping similar data points together or reducing the number of features to make the data easier to understand.
- **Example:**
 - **Customer segmentation:** A business might have data about its customers, like age, income, and purchase history, but without labels. The algorithm groups customers into similar categories (clusters) to target them better.
 - **Market basket analysis:** The algorithm might find patterns in shopping data, like which products are often bought together.

3. Reinforcement Learning:

- **What it is:** Reinforcement learning is where an agent learns to make decisions by interacting with an environment. The agent takes actions and gets **rewards or punishments** based on those actions. The goal is to learn the best strategy (policy) to maximize the reward over time.
- **How it works:** The agent starts by making random decisions, and through feedback (rewards or penalties), it learns which actions lead to better outcomes.
- **Example:**
 - **Game playing:** A computer playing chess or Go learns by playing many games and adjusting its moves based on whether it wins or loses.
 - **Self-driving cars:** The car learns to drive by trying different actions (turning, speeding, stopping) and getting feedback based on whether those actions bring it closer to its goal (safe driving).

To summarize:

- **Supervised Learning:** Learning with labeled data to make predictions (e.g., email classification).
- **Unsupervised Learning:** Finding patterns in unlabeled data (e.g., customer segmentation).
- **Reinforcement Learning:** Learning through trial and error, aiming to maximize rewards (e.g., game playing).

These are the three main types of learning in machine learning, each useful for different kinds of tasks.

Q.8. Describe simple machine learning models such as linear regression and logistic regression

1. Linear Regression:

Linear Regression is a simple machine learning model used for predicting continuous values (like price, temperature, or height) based on input data.

- **Goal:** It tries to find the relationship between input variables (also called features) and the output variable (also called the target).
- **How it works:**
 - Imagine a straight line on a graph. The equation of the line is $y = mx + b$, where:
 - ♦ y is the predicted output (target),
 - ♦ m is the slope (how much the line rises or falls),
 - ♦ x is the input (feature),
 - ♦ b is the y-intercept (where the line crosses the y-axis).
 - The model finds the best line that fits the data points. This is done by minimizing the difference (error) between the predicted values and the actual values in the training data.
- **Example:** Predicting house prices based on the area of the house.

2. Logistic Regression:

Logistic Regression is used for classification tasks, meaning it predicts categories or classes (like "yes/no", "spam/ham", or "cancer/not cancer").

- **Goal:** It predicts a probability that a given input belongs to a certain class.
- **How it works:**
 - Unlike linear regression, logistic regression uses a **sigmoid function** (an S-shaped curve) to map the output to a value between 0 and 1.
 - The equation is: $P = 1 / (1 + e^{(-z)})$, where z is a linear combination of the input features, and e is Euler's number (a constant).
 - The result (P) gives the probability that the data belongs to one class (e.g., "1" or "Yes"). If $P > 0.5$, the model predicts class 1 (or "Yes"), and if $P < 0.5$, it predicts class 0 (or "No").
- **Example:** Predicting whether an email is spam or not based on certain features like keywords.

Key Differences:

- **Linear Regression:** Predicts continuous values (e.g., price, temperature).
- **Logistic Regression:** Predicts probabilities and then classifies the data into categories (e.g., spam or not spam).

These models are quite simple but form the basis for understanding more complex machine learning techniques. Make sure to understand their formulas and how they are used to make predictions!

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Q..9.Explain the difference between supervised learning and unsupervised learning.

1. Supervised Learning:

In **Supervised Learning**, the model is trained using labeled data, meaning that the input data comes with the correct answer (label) attached.

- **How it works:**
 - You provide the algorithm with both the **input data** (features) and the **correct output** (labels). The model learns to map inputs to outputs.
 - After training, the model can make predictions on new, unseen data by using the patterns it learned from the labeled data.
- **Example:**
 - **Classification:** Predicting whether an email is spam or not spam based on features like the subject, sender, and content. The training data includes emails that are already labeled as spam or not spam.
 - **Regression:** Predicting house prices based on features like size, location, and number of rooms, where the price is the label.
- **Key Point:** The model is **supervised** because it gets the correct answer during training.

2. Unsupervised Learning:

In **Unsupervised Learning**, the model is trained using data that **does not have labels**. The goal is to find hidden patterns or structures in the data without knowing the correct answers beforehand.

- **How it works:**
 - You only provide the model with input data (features), and the model tries to learn the underlying structure or relationships within the data.
 - The model might group similar data points together (clustering) or reduce the dimensions of the data (dimensionality reduction).
- **Example:**

- **Clustering:** Grouping customers into different segments based on their purchase behavior, where we don't know in advance which customer belongs to which group.
- **Anomaly Detection:** Finding unusual behavior in network traffic, where the system learns what is "normal" and identifies anything that doesn't fit that pattern.
- **Key Point:** The model is **unsupervised** because it doesn't have labeled answers during training and tries to discover patterns on its own.

Key Differences:

Aspect	Supervised Learning	Unsupervised Learning
Data Type	Requires labeled data (input + correct output)	Uses unlabeled data (only input, no correct output)
Goal	Learn a mapping from input to output (e.g., prediction)	Find patterns or structures in the data (e.g., grouping)
Example	Email classification (spam vs. not spam), house price prediction	Customer segmentation, anomaly detection

In Summary:

- **Supervised Learning** is like a teacher guiding you by showing the correct answers.
- **Unsupervised Learning** is like exploring data on your own without knowing the answers, trying to find hidden patterns.

Make sure to understand the key ideas of each type, as they are the foundation for many machine learning tasks!

Q.10.Discuss the concept of reinforcement learning with an example.

Reinforcement Learning (RL):

Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by interacting with its environment to achieve a goal. The agent receives **rewards** or **penalties** based on the actions it takes, and over time, it learns which actions lead to the best outcomes.

In RL, the agent doesn't know the correct actions initially. Instead, it tries different actions, learns from the results, and improves its decision-making over time.

Key Concepts in Reinforcement Learning:

1. **Agent:** The decision-maker (e.g., a robot, a game character, or a software program).
2. **Environment:** The world the agent interacts with (e.g., a game, a maze, or a robotic system).
3. **Action:** The choice the agent makes (e.g., moving left, jumping, or picking up an object).
4. **State:** The current situation of the agent in the environment (e.g., the position of the agent in a game).
5. **Reward:** Feedback from the environment that tells the agent how good or bad an action was. It can be positive (reward) or negative (penalty).
6. **Policy:** The strategy the agent uses to decide what actions to take in different situations.

How It Works:

- The agent starts in an initial state and performs an action. The environment responds by changing the state and giving the agent a reward (positive or negative).
- The agent uses this feedback (reward or penalty) to improve its actions.
- The goal is for the agent to maximize its total reward over time by learning the best actions (the **optimal policy**).

Example: Training a Robot to Walk:

Imagine a robot in a room trying to learn how to walk toward a target (like a ball) without falling over. Here's how reinforcement learning works in this scenario:

1. **Agent:** The robot.
2. **Environment:** The room where the robot is trying to move.
3. **State:** The robot's current position, angle, and balance in the room.
4. **Action:** The robot can perform actions like moving forward, turning left, or right, or adjusting its position.
5. **Reward:**
 - If the robot gets closer to the target (the ball), it gets a positive reward (e.g., +10).
 - If the robot falls over or moves away from the target, it gets a negative reward (e.g., -10).
6. **Learning:** Over time, the robot tries different actions and learns which ones lead to positive rewards (moving toward the ball) and which ones lead to negative rewards (falling over).

The robot continues to adjust its actions to maximize its rewards and eventually learns to walk toward the target without falling.

Key Points:

- **Trial and Error:** The agent learns by trying different actions and receiving feedback. It doesn't know the best actions at the beginning.
- **Delayed Reward:** The reward may not be immediate. For example, the robot might get a positive reward only after successfully reaching the target.
- **Exploration vs. Exploitation:** The agent has to balance exploring new actions (trying something unknown) and exploiting the actions it knows work well.

In Summary:

Reinforcement Learning is like teaching an agent through rewards and penalties. The agent tries different actions, learns from the results, and gradually becomes better at making decisions to achieve a goal.

I hope this explanation makes it clear! Reinforcement learning is powerful and is used in many real-world applications, like game AI, robotics, and even self-driving cars.

Q.11. Discuss the forms of learning in machine learning: supervised learning, unsupervised learning, and reinforcement learning. Provide examples and explain how each form works.

1. Supervised Learning:

Supervised Learning is a type of machine learning where the model learns from labeled data. This means the input data has corresponding output labels, and the algorithm learns to map the inputs to the correct outputs.

How it works:

- You train the model with **input-output pairs** (labeled data).
- The model learns the relationship between the inputs and their corresponding outputs.
- Once trained, the model can predict the output for new, unseen data.

Example:

- **Email Classification:** Predicting whether an email is spam or not spam. The training data consists of emails labeled as "spam" or "not spam." The model learns to classify emails based on features like the sender, subject, and keywords.

- **House Price Prediction:** Predicting the price of a house based on features like size, location, and number of rooms. The training data includes houses with known prices (labels).

Key Points:

- **Labeled data** is used.
- The model **learns from the correct answers** (labels).
- Common algorithms: **Linear Regression, Logistic Regression, Decision Trees, Support Vector Machines (SVM).**

2. Unsupervised Learning:

Unsupervised Learning is when the model learns from **unlabeled data**, meaning there are no correct answers provided. The goal is to find hidden patterns, relationships, or structures in the data.

How it works:

- The model works with data that has no labels.
- It tries to find structure or patterns in the data, such as grouping similar data points together (clustering) or reducing the number of features (dimensionality reduction).

Example:

- **Customer Segmentation:** In marketing, businesses might use unsupervised learning to group customers with similar behaviors. For example, customers might be grouped based on their buying habits without any predefined labels.
- **Anomaly Detection:** Detecting unusual patterns in data, such as fraud detection in credit card transactions, where the model identifies transactions that don't fit normal patterns.

Key Points:

- **Unlabeled data** is used.
- The model tries to **discover hidden patterns** or structures.
- Common algorithms: **K-means Clustering, Principal Component Analysis (PCA), Hierarchical Clustering.**

3. Reinforcement Learning:

Reinforcement Learning (RL) is a type of machine learning where an **agent learns to make decisions** by interacting with an environment and receiving feedback in the form of rewards or penalties.

How it works:

- The agent starts with little or no knowledge of the environment.
- It takes actions in the environment, and based on its actions, it receives **rewards** (positive feedback) or **penalties** (negative feedback).
- Over time, the agent learns to take actions that maximize its total reward (this is called the **optimal policy**).

Example:

- **Game Playing:** Imagine an AI playing a game like Chess or Go. The AI agent tries different moves and receives feedback (winning or losing). It learns which moves are best to win the game by maximizing its rewards.
- **Robot Navigation:** A robot in a maze learns to reach the target by taking different actions (turning, moving forward) and getting rewards when it gets closer to the target, or penalties when it hits obstacles.

Key Points:

- The model learns through **trial and error**.
- The agent learns from **rewards** and **penalties** to improve its actions.
- Common algorithms: **Q-learning, Deep Q Networks (DQN), Monte Carlo Methods**.

Comparison Table:

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Type	Labeled data (input + output)	Unlabeled data (only input)	No predefined labels; learns by interacting with the environment
Goal	Learn a mapping from inputs to outputs	Discover hidden patterns or structures	Learn the best actions to maximize reward over time
Examples	Spam detection, House price prediction	Customer segmentation, Anomaly detection	Game AI, Robot navigation, Self-driving cars
Algorithms	Linear Regression, Logistic Regression, SVM	K-means Clustering, PCA, Hierarchical Clustering	Q-learning, Deep Q Networks, Monte Carlo Methods

In Summary:

1. **Supervised Learning:** The model learns from labeled data (input + output) to predict outcomes (e.g., email classification).
2. **Unsupervised Learning:** The model learns from unlabeled data to find patterns or groupings (e.g., customer segmentation).
3. **Reinforcement Learning:** The agent learns to make decisions based on rewards and penalties through trial and error (e.g., game AI or robot navigation).

Each of these learning types has its own use cases, and understanding their

differences will help you choose the right approach for different machine learning tasks.

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Q.12. Explain simple machine learning models such as linear regression, logistic regression, and support vector machines (SVM). How are these models applied in real-world problems?

1. Linear Regression:

Linear Regression is a simple machine learning model used for predicting a continuous value (like price, temperature, etc.) based on input features (like size, area, etc.).

How it works:

- The model tries to find the relationship between the input features (like the size of a house) and the output (like the price of the house).
- It does this by fitting a **straight line** (linear relationship) to the data.
- The formula is:
 - $y = mx + b$, where:
 - ♦ y is the predicted value (e.g., house price),
 - ♦ m is the slope of the line (how much the value increases or decreases),
 - ♦ x is the input feature (e.g., size of the house),
 - ♦ b is the y-intercept (where the line crosses the y-axis).

Example in Real-World:

- **Predicting House Prices:** If we have data about houses (like the number of rooms, size, and location), linear regression can predict the price of a house based on these features.
- **Sales Forecasting:** Predicting future sales of a product based on historical sales data.

2. Logistic Regression:

Logistic Regression is a machine learning model used for **classification tasks**, where the goal is to predict a category or class (e.g., yes/no, spam/not spam).

How it works:

- Unlike linear regression, **logistic regression** predicts **probabilities** that a given input belongs to a certain class (e.g., the probability that an email is spam).
- It uses the **sigmoid function** to map the output to a probability between

0 and 1. The formula is:

- $P = 1 / (1 + e^{-z})$, where:
 - ♦ **P** is the probability of the positive class,
 - ♦ **z** is a linear combination of input features (like weight, size, etc.),
 - ♦ **e** is Euler's number (a constant).
- If **P > 0.5**, the model predicts one class (e.g., "Spam"), and if **P < 0.5**, it predicts the other class (e.g., "Not Spam").

Example in Real-World:

- **Email Spam Detection:** Logistic regression can classify whether an email is spam or not based on certain features, like keywords and sender.
- **Medical Diagnosis:** Predicting whether a patient has a disease based on medical features (e.g., age, blood pressure).

3. Support Vector Machines (SVM):

Support Vector Machines (SVM) is a more advanced machine learning algorithm used for **classification** and sometimes for **regression**. It finds the best boundary (called a hyperplane) that separates the data into different classes.

How it works:

- The goal of SVM is to **maximize the margin** between the classes. This means it tries to find the line or plane that **best divides** the data into two categories.
- SVM is good for cases where data is not linearly separable, meaning it cannot be separated by a straight line. It uses techniques like the **kernel trick** to handle complex, non-linear data by mapping it to a higher-dimensional space.
- SVM tries to find the **support vectors**, which are the data points that are closest to the hyperplane.

Example in Real-World:

- **Image Classification:** SVM is widely used in image classification tasks, like recognizing handwritten digits (e.g., recognizing digits in zip codes).
- **Face Detection:** In security systems, SVM can be used to classify whether an image contains a human face or not.

Key Differences Between These Models:

Model	Type of Problem	Prediction Type	Real-World Examples
Linear Regression	Predicting a continuous value	Regression (continuous)	Predicting house prices, sales forecasting

Logistic Regression	Classifying into categories (yes/no)	Classification (binary)	Spam detection, medical diagnosis
Support Vector Machines (SVM)	Classifying data into categories	Classification (binary or multi-class)	Image classification, face detection, email classification

In Summary:

1. **Linear Regression** is used for predicting continuous values, like prices or temperatures. It's based on finding the best straight line that fits the data.
 2. **Logistic Regression** is used for binary classification problems, like predicting whether an email is spam or not, by estimating the probability that something belongs to a class.
 3. **Support Vector Machines (SVM)** is a powerful classifier that finds the best boundary to separate different classes in the data, and it's great for complex classification tasks, especially when data is not easily separable.
- These models are widely used in real-world applications like email filtering, medical diagnosis, sales forecasting, and even self-driving cars!